Open Source Software Capstone Project Report

Open Source Software Audit using Linux Shell Scripting

Student Details

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Course: Open Source Software
Project Title: OSS Audit Project
Tool Selected: Git
Environment: Linux (Ubuntu / GitHub Codespaces)

Abstract

Open Source Software (OSS) plays a significant role in modern computing and software development. It allows users to access, modify, and distribute software freely. This project focuses on auditing an open-source environment using Linux shell scripting and Git. The main objective of the project is to understand the working of Linux systems, open-source tools, and automation through shell scripts.

The project includes five shell scripts that perform system identity analysis, package inspection, disk auditing, log analysis, and manifesto generation. Git is selected as the open-source tool for analysis due to its wide usage in version control and collaborative development.

The scripts were executed in a Linux environment using GitHub Codespaces, and outputs were generated successfully. The project demonstrates the importance of open-source software, Linux command-line utilities, and automation techniques in system auditing.

This project helps in understanding open-source philosophy, system-level scripting, and Git-based project management in a practical way.

Introduction

Open Source Software refers to software whose source code is freely available to users for modification and distribution. It promotes transparency, collaboration, and innovation in software development. Open-source tools are widely used in operating systems, web development, data science, cloud computing, and system administration.

Linux is one of the most popular open-source operating systems. It provides powerful command-line utilities and scripting capabilities that help in automation and system management. Shell scripting in Linux allows users to create automated scripts for system auditing, log analysis, disk inspection, and software management.

Git is a distributed version control system used to track changes in source code and manage collaborative development. It is open-source and widely used by developers across the world.

This project focuses on creating shell scripts that audit system information and demonstrate the practical use of open-source tools in Linux environments.

Objectives

The main objectives of this project are:

- To understand Open Source Software concepts
- To explore Linux system environment
- To learn Git and version control
- To create shell scripts for system auditing
- To analyze disk and log files
- To automate system information retrieval
- To generate an open-source manifesto
- To maintain project using GitHub

Tool Selection

Selected Tool: Git

Git is a distributed version control system designed to track changes in source code and coordinate work among multiple developers. It is fast, secure, and efficient.

Git is widely used in software development, open-source projects, and DevOps environments.

Features of Git

- Distributed version control system
- Open-source and free
- Fast and efficient
- Supports collaboration
- Tracks project history

- Easy branching and merging
- Secure and reliable

License

Git is released under the **GNU General Public License (GPL)**.

Technologies Used

- Linux (Ubuntu Environment)
- Bash Shell Scripting
- Git
- GitHub
- Linux Commands
- GitHub Codespaces

System Environment

The project was executed in a Linux environment using GitHub Codespaces.

Environment Details

- Operating System: Ubuntu Linux
- Shell: Bash
- Tool: Git
- Platform: GitHub Codespaces
- Terminal: Linux Terminal

Script 1: System Identity

Description

This script displays system identity information. It helps in understanding the Linux system configuration and environment.

Functions

- Displays Linux distribution
- Shows kernel version
- Displays current user
- Shows system uptime
- Displays date and time
- Shows GPL license information

Command

```
None
./script1_system_identity.sh
```

Output

```
@2608raghav →/workspaces/oss-audit-24BCE10427 (main) $ chmod +x *.sh
@2608raghav →/workspaces/oss-audit-24BCE10427 (main) $ ./script1_system_identity.sh
======================================
 Open Source Audit — Raghav Gupta
======================================
Software Selected : Git
Linux Distribution: Ubuntu 24.04.3 LTS
Kernel Version    : 6.8.0-1044-azure
Current User      : codespace
System Uptime     : up 5 minutes
Current Date/Time : Fri Mar 27 19:47:20 UTC 2026
Linux is covered under GPL License
```

Script 2: Package Inspector

Description

This script checks whether Git is installed in the system and displays its version and description.

Functions

- Checks Git installation
- Displays Git version
- Shows Git description

Command

None

```
./script2_package_inspector.sh
```

Output

```
@2608raghav →/workspaces/oss-audit-24BCE10427 (main) $ ./script2_package_inspector.sh
git is installed
git version 2.52.0
Git: Distributed version control system
```

Script 3: Disk Auditor

Description

This script audits important system directories and displays permissions, owner, and size.

Directories Checked

- /etc
- /var/log
- /home
- /usr/bin
- /tmp

Command

None

```
./script3_disk_auditor.sh
```

Output

```
@2608raghav →/workspaces/oss-audit-24BCE10427 (main) $ ./script3_disk_auditor.sh
Directory Audit Report
/etc => drwxr-xr-x root root | Size: 3.3M
/var/log => drwxr-xr-x root root | Size: 824K
/home => drwxr-xr-x root root | Size: 283M
/usr/bin => drwxr-xr-x root root | Size: 534M
/tmp => drwxr-xrwt+ root root | Size: 52K
```

Script 4: Log Analyzer

Description

This script analyzes system log files and searches for specific keywords.

Functions

- Accepts log file as input
- Accepts keyword
- Searches keyword using grep
- Displays occurrences

Command

```
None
./script4_log_analyzer.sh /var/log/dpkg.log error
```

Output

```
@2608raghav ➜ /workspaces/oss-audit-24BCE10427 (main) $ ./script4_log_analyzer.sh /var/log/dpkg.log error
Keyword 'error' found 48 times
2025-11-27 10:33:11 status unpacked liberror-perl:all 0.17029-2
2025-11-27 10:33:13 configure liberror-perl:all 0.17029-2 <none>
2025-11-27 10:33:13 status unpacked liberror-perl:all 0.17029-2
2025-11-27 10:33:13 status half-configured liberror-perl:all 0.17029-2
2025-11-27 10:33:13 status installed liberror-perl:all 0.17029-2
```

Script 5: Manifesto Generator

Description

This script generates an open-source manifesto based on user input.

Functions

- Takes user input
- Generates manifesto
- Stores output in manifesto.txt

Command

```
None
./script5_manifesto_generator.sh
```

Example Input

```
None
Tool: git
Freedom means: sharing
Build: open source software
```

Output

```
@2608raghav ➜ /workspaces/oss-audit-24BCE10427 (main) $ ./script5_manifesto_generator.sh
Answer questions
Tool: git
Freedom means: sharing
Build: open source software
On Fri Mar 27 19:49:36 UTC 2026 I believe freedom is sharing.
Tools like git inspire innovation.
I want to build open source software and share it.
```

GitHub Repository

Repository Link:

https://github.com/2608raghav/oss-audit-24BCE10427

The repository contains:

- Shell scripts
- README file
- Screenshots
- Project report

Learning Outcomes

- Understanding of Open Source Software
- Linux shell scripting knowledge
- Git and GitHub usage
- System auditing techniques
- Log file analysis
- Disk inspection
- Automation using Bash
- Open-source philosophy

Advantages of Open Source Software

- Free to use
- Transparent
- Secure
- Flexible
- Community support
- Encourages innovation
- Easy collaboration

Challenges Faced

- Linux environment setup
- WSL installation issues
- Log file availability
- Script execution errors
- GitHub configuration

These challenges were resolved using GitHub Codespaces and Linux terminal.

Conclusion

The OSS Capstone Project successfully demonstrates the practical implementation of open-source concepts using Linux and Git. The shell scripts were developed to audit system information, analyze logs, inspect disk directories, and generate an open-source manifesto.

The project highlights the importance of open-source tools in modern computing and shows how Linux scripting and Git can be used to automate system-level tasks efficiently.

This project helped in gaining practical knowledge of Linux, Git, shell scripting, and open-source philosophy.

References

- https://git-scm.com
- https://opensource.org
- https://www.gnu.org
- https://ubuntu.com
- https://linux.org

Author

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